Reducing Customer Churn at Meridian Retail Group: An End-to-End Data Science Investigation.

Presented by

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1. Problem Statement

Meridian Retail Group, a mid-sized multi-channel retailer, faces rising customer churn. Retaining customers is critical for profitability, as acquiring new customers is significantly more expensive than keeping existing ones. The business challenge is to diagnose the drivers of churn and build a predictive model that identifies at-risk customers, enabling proactive retention strategies.

2. Methodology

- **Data Collection & Cleaning**: Synthetic retail data (customers, orders, products, support tickets) was cleaned to remove duplicates, fix inconsistent formats, and handle missing values.
- **Exploratory Data Analysis (EDA)**: Visualizations were used to understand churn distribution, customer demographics, purchase behavior, and support interactions.
- **Feature Engineering**: Key features included recency, frequency, monetary value (RFM), support ticket count, average satisfaction, and resolution time.
- **Model Development**: Multiple machine learning models were tested (Logistic Regression, Random Forest, Gradient Boosting).
- **Validation**: Models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.
- **Deployment**: A Flask API was built to serve predictions, with robust input validation for missing fields, invalid types, and categorical errors.

3. Findings

- **Recency**: Customers who had not purchased recently were more likely to churn, (Customers inactive for >365 days are 3x more likely to churn).
- **Frequency**: Low purchase frequency strongly correlated with churn.

- **Support Experience**: High resolution times and low satisfaction scores increased churn risk.
- **Monetary Value**: High spenders are less likely to churn, but dissatisfied high spenders churn faster
- **Membership Tier**: Platinum members showed the lowest churn rates, while Silver members were more vulnerable.
- **Preferred Channel**: Online customers had slightly higher churn compared to mobile app users, suggesting engagement differences.

4. Model Results

- **Gradient Boosting** delivered the best performance:
    - Accuracy: ~85%
    - Precision: ~82%
    - Recall: ~79%
    - ROC-AUC: ~0.91
- Logistic Regression and Random Forest performed reasonably but were less effective in capturing non-linear relationships.

5. Business Recommendations

1. **Retention Campaigns**: Target customers with low frequency and high recency scores with personalized offers.
2. **Customer Support Improvements**: Reduce resolution times and improve satisfaction scores through better training and automation.
3. **Loyalty Programs**: Strengthen incentives for Silver-tier customers to upgrade to Gold/Platinum.
4. **Channel Engagement**: Enhance online shopping experience to reduce churn compared to mobile app users.
5. **Proactive Monitoring**: Use the API to flag at-risk customers in real time and trigger retention workflows.

6. Limitations

- **Synthetic Data**: The dataset was generated and may not fully reflect real-world customer behavior.
- **Feature Scope**: Limited to available attributes; external factors (competitor actions, macroeconomic trends) were not included.
- **Model Bias**: Predictions depend on historical patterns; unseen behaviors may reduce accuracy.
- **Deployment Constraints**: Current API is a prototype; production deployment would require scaling, monitoring, and integration with CRM systems.

7. Conclusion

This project demonstrates how data science can diagnose churn drivers and predict at-risk customers. By combining business insights with machine learning, Meridian Retail Group can reduce churn, improve customer satisfaction, and increase profitability. The deployed API provides a practical tool for integrating predictive analytics into business operations.